

Operative records searched for a diagnosis naming intestinal malrotation or a Ladd procedure, 1 Dec 2012 - 30 Jun 2026
559 records in 499 children, all reviewed

Surgically confirmed intestinal malrotation
503 records in n = 465 children (reference standard)

At least one preoperative index test
n = 410 (740 index examinations)

UGI series 301 Abdominal CT 320
Gastrointestinal ultrasound 119
All three modalities 59

Index unit = examination episode closest to operation
812 eligible preoperative reports → 740 episodes
(same-day reports of one modality pooled; 34 earlier repeat examinations not audited)

Each episode classified as positive or negative by rule-based algorithm + surgeon adjudication, and separately audited for documented technical content

None of the three index tests
n = 55 (11.8%)

but every one had other preoperative imaging:

abdominal / chest radiograph	46
contrast enema of the colon	12
ultrasound of another region	22
CT of another region	7
no in-hospital study	5

(all 5 with documented outside imaging)

Outside or outpatient imaging documented in 33;
already reporting malrotation or volvulus in 19

*Detection rates are computed among children with surgically confirmed malrotation who received the test.
The cohort contains no test-negative children, so specificity and predictive values are not estimable.*